

Examiner
only

4. (a) Complete the following sentences about genes.

[2]

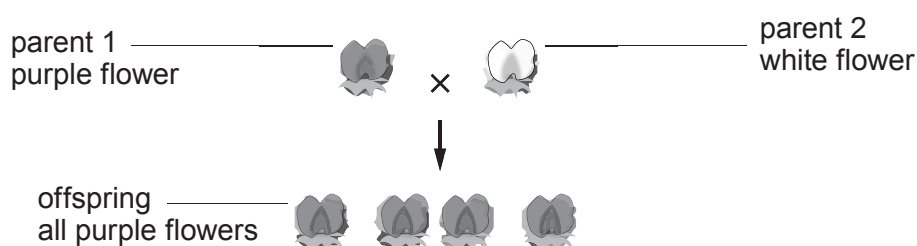
- (i) Genes are found in the nucleus of a cell on paired structures called

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- (ii) Genes are sections of a long molecule called

- (b) Alan crossed two pea plants (*Pisum sativum*).

- Parent 1 had purple flowers.
- Parent 2 had white flowers.
- All the offspring had purple flowers, as shown below.



Flower colour in pea plants is controlled by one gene. The gene has two alleles, **D** and **d**.

- (i) Underline the correct answer to complete the following sentences.

- I. The genotype of parent 1 is **DD / Dd / dd** [1]

- II. The genotype of parent 2 is **DD / Dd / dd** [1]

- (ii) State the phenotype of the offspring. [1]

.....

- (iii) The offspring are heterozygous.

State the meaning of the term *heterozygous*. [2]

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